Design and Deployment of an Imaging Spectrometer for the Characterisation of Spatial Chirp

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Introduction

1.1 Plasma Modulated Plasma Accelerator

A plasma can be formed by ionising a gas target using a high-intensity, short (10 fs) laser pulse. The interaction of the laser pulse with the plasma can form accelerating structures with field strengths of the order of 100 GV/m. In a laser-wakefield accelerator scheme (LWFA), these accelerating structures may be used to achieve electron energies in the order of 100s MeV to GeV [1]. Such schemes may lead to the production of compact, low cost electron sources with applications such as light source generation [2], QED experiments [3], radiobiology studies [4], and probing fundamental plasma physics [5].

LWFA schemes are currently limited by the repititon rates of their laser systems, most commonly $\lessapprox$ 10 Hz with Ti:sapphire lasers. The University of Oxford's Laser Plasma Accelerator Group (LPAG) is working with collaborators on the plasma-modulated plasma accelerator (P-MoPA) project to address this [6, 7, 8, 9, 10]. The project aims to use thin-disk lasers with high repititon rates ($\sim$ 1 kHz) to drive a LWFA electron source. The pulse duration of such lasers is too long too drive a LWFA, so the following scheme is used to create a pulse-train (see Figure 1.1): A short, low-energy seed-pulse is co-propagated with a long drive-pulse through a modulator stage containing a gas. The seed-pulse ionises the gas and drives a low-intensity plasma wave that modulates the drive-pulse, producing side-bands in its spectrum. A compressor then removes the phase on these side-bands, converting the picosecond pulse into a train of short pulses spaced temporally by the plasma period. This pulse train can then be directed into a second plasma stage of the same density, resonantly exciting a wakefield into which electrons can be injected and subsequently accelerated. This modulation-compression process is sensitive to the drive-pulse's spatio-spectral content; any spatio-spectral couplings would vary the modulation across the pulse's

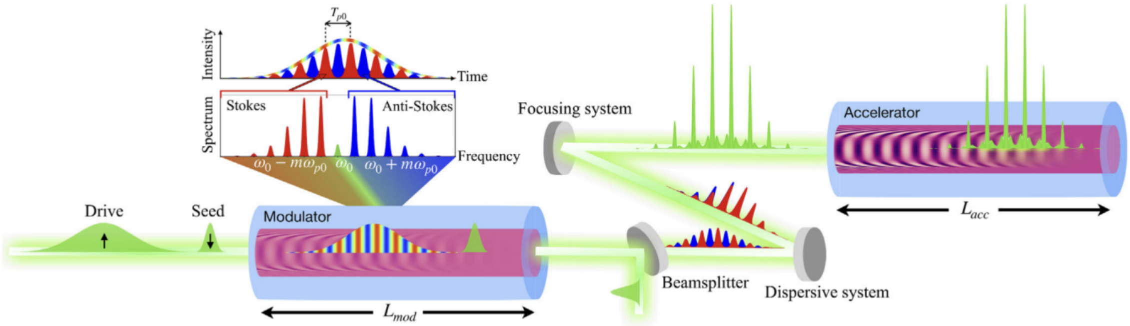


Figure 1.1: The Plasma Modulated Plasma Accelerator (P-MoPA) scheme. Reproduced from [6].

profile and degrade the resulting pulse-train, weakening the pulse-train's ability to drive a wakefield and leading to unstable electron acceleration. The characterisation of spatio-spectral couplings in the thin-disk lasers used in the P-MoPA project is thus essential for the project's success.

1.2 Angular and Spatial Chirp

1.2.1 Chirped Pulse Amplification

Prior to Gérard Mourou and Donna Strickland's invention of chirped pulse amplification (CPA) [11], for which they were awarded the Nobel Prize in Physics in 2018, it was extremely challenging to produce ultrashort laser pulses with enough intensity for modern LWFA without damaging the optical equipment and introducing non-linear effects. CPA avoids these issues through a staged amplification procedure (see Figure 1.2): An ultrashort, low-amplitude pulse enters a pulse stretcher, composed of two identical parallel diffraction gratings. This stretches the pulse temporally, reducing its intensity without loss of energy. The pulse is subsequently amplified then directed through a pulse compressor, similar to the pulse stretcher, which compresses the pulse back to its original duration. The result is a high-intensity ultrashort pulse, but crucially this intensity is only reached post-compressor, leaving all previous optical components unharmed.

As with most real optical systems, CPA systems are highly sensitive to misalignments. In particular, a relative misalignment between the two gratings of the stretcher or the compressor, or a mismatch between the dispersive properties of the stretcher and the compressor, may produce spatio-spectral couplings in the pulse. One such coupling is known as *angular chirp*, a correlation between the angle of propagation

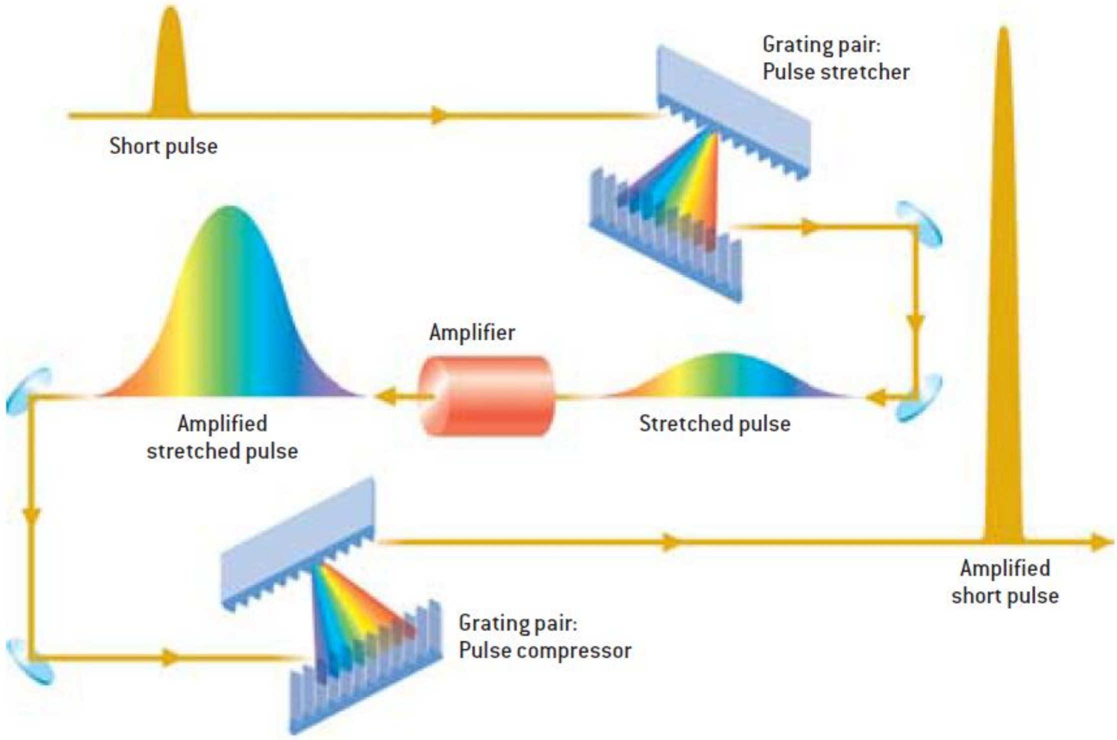


Figure 1.2: Illustration of Chirped Pulse Amplification (CPA). Reproduced from [12].

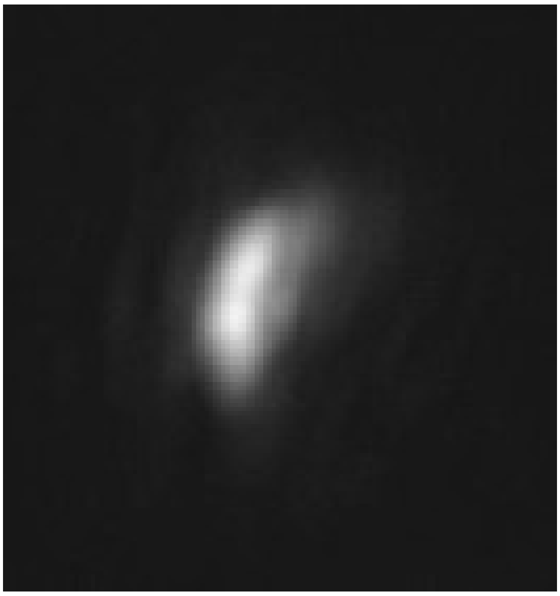
θ and wavelength λ within a pulse of finite bandwidth. When an angularly chirped pulse propagates over a finite distance, it acquires *spatial chirp*, a correlation between position x along an arbitrary transverse axis across a pulse and wavelength λ.

1.2.2 Angular-to-Spatial Chirp Conversion at Focus

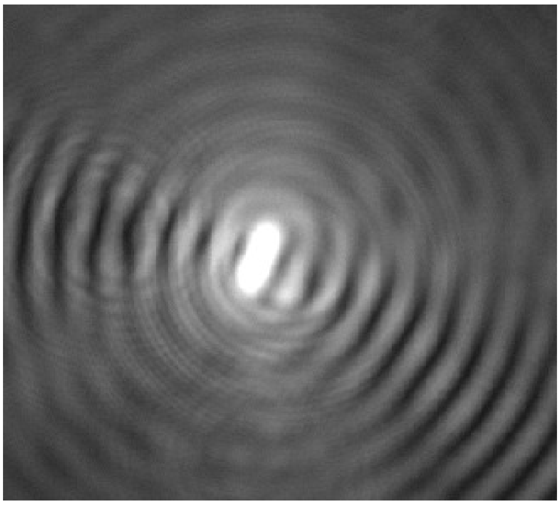
A useful property of the converging lens is the conversion between angular and spatial information at its focal plane. In particular, any collimated plane wave incident on an ideal thin lens of focal length f is, in the paraxial approximation, focussed to a transverse position x on the lens' back focal plane proportional to its angle of incidence θ:

$$x = f\theta. \tag{1.1}$$

Consequently, if the incident pulse is angularly chirped, meaning θ is spectrally dependent, the transverse position x of the resulting focal spot also becomes spectrally dependent, meaning it is spatially chirped. The result is a focal spot that is stretched along an arbitrary transverse axis, as opposed to a typical circular focal spot. (see Figure 1.3).



(a) Lens focus



(b) Axicon focus

Figure 1.3: Spatially chirped foci resulting from angularly chirped incident pulses. Images courtesy of Alexander Harrison.

Differentiating Equation 1.1 with respect to wavelength λ:

$$\frac{dx}{d\lambda} = f\frac{d\theta}{d\lambda}. \tag{1.2}$$

The derivative $\zeta \equiv \frac{dx}{d\lambda}$ is known as *spatial dispersion*, corresponding to a pulse's spectrally dependent transverse position $x(\lambda)$, and is the conventional metric used for describing spatial chirp throughout this project. Another common metric used for describing spatial chirp is known as the frequency gradient, $\frac{d\omega}{dx}$, corresponding to a pulse's transverse-positionally dependent angular frequency $\omega(x)$. While these parametarisations both describe spatial chirp, they are not simply reciprocals of each other [13]. Spatial dispersion is used throughout this project because of its convenient relationship with the angular dispersion $\frac{d\theta}{d\lambda}$ that produces it. Wavelength is chosen as the spectral unit throughout this project due to its direct relationship with the diffraction angle of optical gratings.

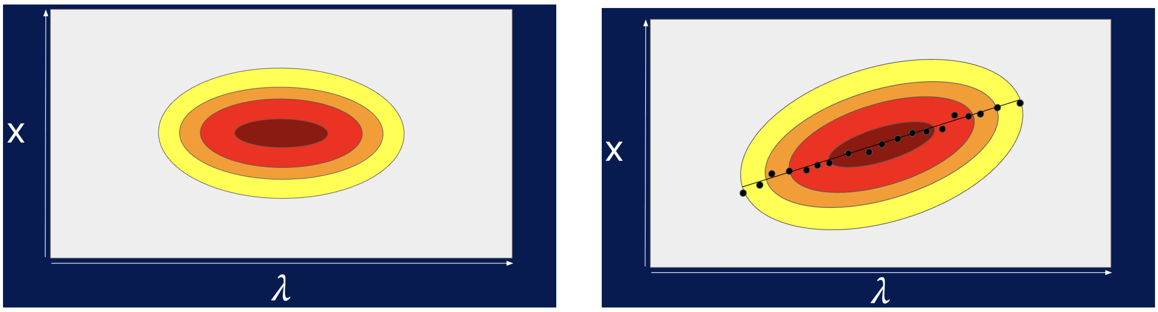
1.2.3 Consequences for P-MoPA

A spatially chirped focal spot occupies a larger area than otherwise, reducing the focus' peak intensity. Furthermore, the spatial seperation of optical frequencies at the focus means the local bandwidth is reduced, which can lengthen the pulse's duration and also resulting in a reduced peak intensity. For a channel-forming pulse used to ionise a plasma channel for the P-MoPA accelerator stage, a sufficient reduction in peak intensity may lead to incomplete ionisation or a lower ionisation state, modifying the electron-density profile of the plasma formed [14]. If this changes the effective plasma density experienced by the pulse train, then this is particularly detrimental to the P-MoPA scheme because the modulation and acceleration plasma stages must share the same plasma frequency. This is precisely what allows the pulse train to have the appropriate spacing required to resonantly drive a wakefield in the accelerator stage [8]. Additionally, the initially ionised plasma column may have a smaller radius, as the radial distance at which the pulse's transverse intensity distribution crosses the ionisation threshold moves inward. At sufficiently low intensity, the volume of gas ionised and the energy deposited into the plasma may become insufficient to form a guiding channel with the desired properties [14].

The wakefield is driven by the ponderomotive force associated with the pulse train, which is proportional to the gradient of the laser intensity [1, 6]. If the pulse train is focussed to a spatially chirped focal spot at the entrance to the accelerator stage's plasma channel, each pulse's peak intensity is reducd and its intensity distribution broadened, generally reducing the intensity gradients, leading to a wakefield with reduced field strength. Electrons are therefore accelerated to lower energies, reducing the utility of the project for its various applications [9]. Furthermore, if a pulse's frequency components enter the plasma channel at varying transverse positions and angles due to spatial and angular chirp, they may not all couple equally into the desired guiding mode, reduced the fraction of energy coupled into this mode. Higher order modes may also be excited, causing asymmetric transverse field distributions, and producing interference that oscillates the intensity distribution longitudinally down the plasma channel [9, 15].

Aside from spatial chirp at the focus, angular chirp is also closely associated with pulse-front tilt, where different transverse positions reach peak intensity at different times [16]. This spatio-temporal coupling leads to a transversely asymmetric drive of the plasma wave, producing an asymmetric wakefield that may deflect accelerated electrons from the optical axis [17]. While this effect may be used to steer the accelerated electron beam, it is not always desirable, and the ability to control it is essential.

Within the modulator, angular chirp may also reduce the coupling of the seed and drive pulses to the plasma channel's guided mode as well as modifying their spatial overlap with eachother, distorting the modulation of the drive pulse. These



(a) $\zeta = 0$ (b) $\zeta \neq 0$

Figure 1.4: Illustration of imaging spectrometer-camera images of an un-spatially chirped pulse (a) and a spatially chirped pulse with plotted $x_0(\lambda)$ and a line of best fit (b).

distortions may affect the pulse train and can cause further issues in the accelerator stage.

1.3 Measuring Spatial Chirp

One must be able to detect the existance of spatio-spectral couplings in a laser system before being able to correct them. A common device used for such detection is called an imaging spectrometer [13]. It works by selecting a thin transverse slice of a pulse's spatial profile, before dispersing it in the orthogonal transverse direction. The result is a two-dimensional image with one axis corresponding to wavelength λ, and the other corresponding to position x along a transverse axis across the pulse's spatial profile (see Figure 1.4a). One can then observe the variation of a given wavelength's intensity across the beam's spatial profile, and identify the centroid position x_0, representing the center of that wavelength's intensity distribution. Repeating this for a discretised set of wavelengths results in a dataset for $x_0(\lambda)$, to which various models can be fit to characterise the spatio-spectral coupling (see Figure 1.4b). Differentiating these fitted models allows for computing the pulse's spatial dispersion ζ.

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9.1 Conclusions

9.2 Future Work

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